

Final Project Report

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Data Analysis

```
library(ggplot2)
library(tidyverse)
library(refund)
library(gridExtra)
library(patchwork)

fires <- vroom::vroom("forestfires.csv")
```

Modeling

Regular Model

```
FullModel <- pffr(
  Y_log ~ ff(X_log, xind = s),
  yind = t,
  method = "NCV",
  bs.int = list(bs = "ps", k = 5, m = c(2, 1))
)
FullBetas <- coef(FullModel, n1 = 12, n2 = 12)$smterms[[2]]$value
```

using seWithMean for s(t.vec) .

```
R2_reg <- summary(FullModel)
# R-squared of .568
```

Historical Model

```
FullModel_Hist <- pffr(  
  Y_log ~ ff(X_log, xind = s, limits = 's<t'),  
  yind = t,  
  method = "NCV",  
  bs.int = list(bs = "ps", k = 5, m = c(2, 1))  
)  
FullHistBetas <- coef(FullModel_Hist, n1 = 12, n2 = 12)$smterms[[2]]$value
```

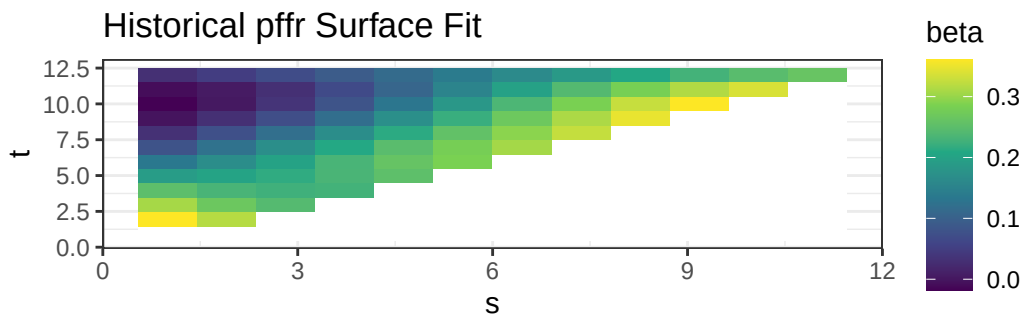
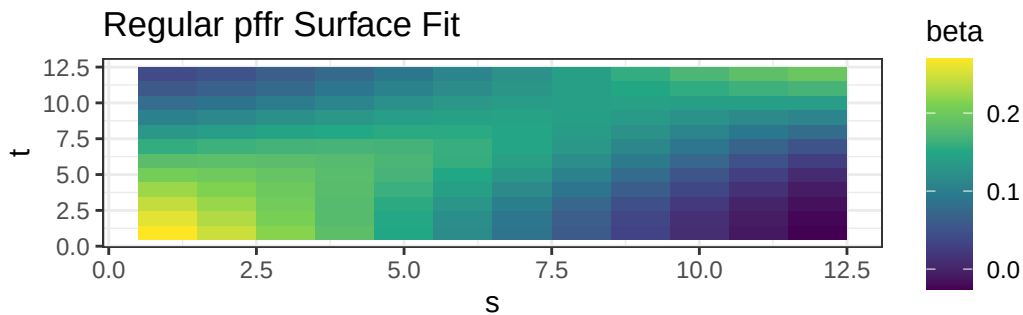
using seWithMean for $s(t.vec)$.

```
R2_hist <- summary(FullModel_Hist)  
# R-squared of .6
```

Plotting the Models

using seWithMean for $s(t.vec)$.

using seWithMean for $s(t.vec)$.



Predictions

Regular Train Model

```
Train_Model <- pffr(  
  Y_log_train ~ ff(X_log_train, xind = s),  
  yind = t,  
  method = "NCV",  
  bs.int = list(bs = "ps", k = 5, m = c(2, 1))  
)  
Train_FullBetas <- coef(Train_Model, n1 = 12, n2 = 12)$smterms[[2]]$value
```

using seWithMean for s(t.vec) .

```
R2_full_train <- summary(Train_Model)  
# R^2 = 0.613
```

Historical Train Model

```
Train_Model_Hist <- pffr(  
  Y_log_train ~ ff(X_log_train, xind = s, limits = 's<t'),  
  yind = t,  
  method = "NCV",  
  bs.int = list(bs = "ps", k = 5, m = c(2, 1))  
)  
Train_HistBetas <- coef(Train_Model_Hist, n1 = 12, n2 = 12)$smterms[[2]]$value
```

using seWithMean for s(t.vec) .

```
R2_hist_train <- summary(Train_Model_Hist)  
# R-squared of .637
```

Regular Model Prediction

```

X_log_test <- log(X_fires_test + 1)
reg.train_preds <- predict(
  Train_Model,
  newdata = list(X_log_train = X_log_test)
)
# Look at RMSE
y_test <- t(Y_area_test)
reg.train_preds_t <- t(reg.train_preds)
regular_rmse <- sqrt(mean((y_test - reg.train_preds_t)^2))
regular_rmse

```

```
[1] 141.591
```

Historical Model Prediction

```

## We need to do this manually
## Predict function doesn't work with limits
# Get intercept
smterms <- coef(Train_Model_Hist, n1 = 12, n2 = 12)$smterms[[1]]$value

```

using seWithMean for s(t.vec) .

```
pters <- coef(Train_Model_Hist, n1 = 12, n2 = 12)$pters[1, 1]
```

using seWithMean for s(t.vec) .

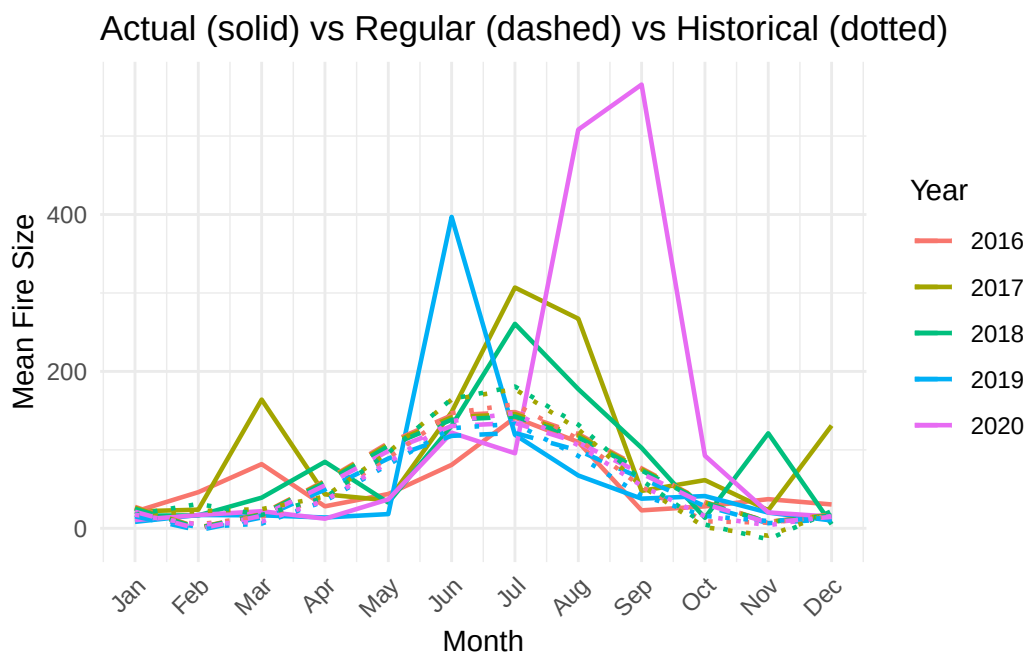
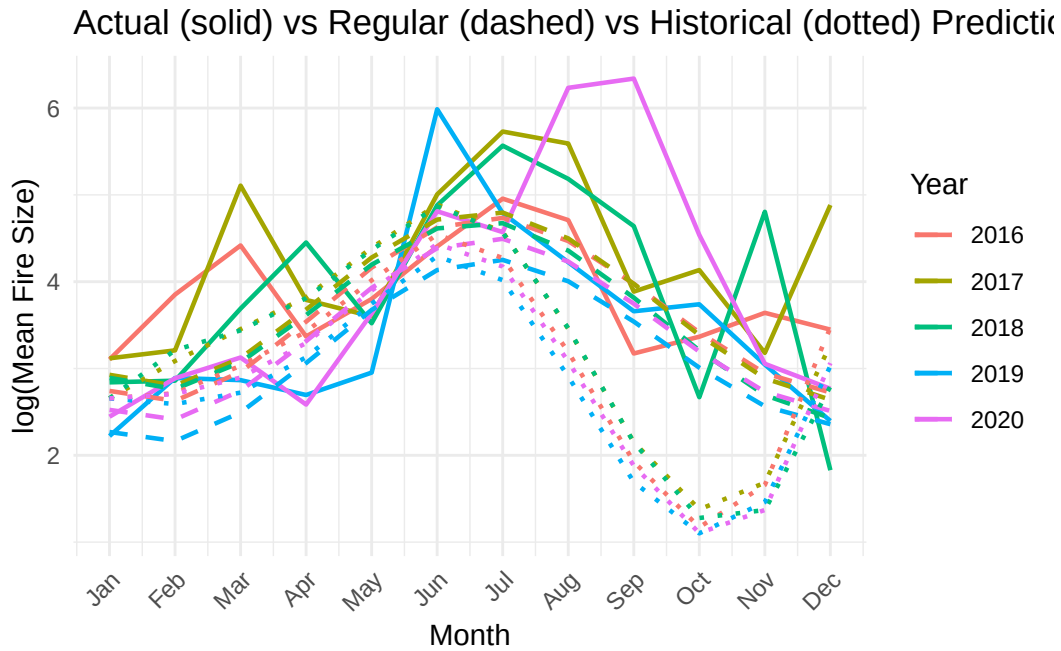
```

b0 <- pters + smterms
b0_matrix <- matrix(b0, nrow = 12, ncol = 5)
# Get the beta surface matrix (12 x 12)
b1 <- Train_HistBetas
b1 <- matrix(b1, nrow = 12, ncol = 12)
b1[lower.tri(b1, diag = TRUE)] <- 0 # taking out values where s>=t
# Manual prediction
b1_matrix <- X_log_test %*% b1
b1_matrix_t <- t(b1_matrix)
hist.train_preds <- b0_matrix + b1_matrix_t
## Get RMSE's
historical_rmse <- sqrt(mean((y_test - hist.train_preds)^2))
historical_rmse

```

[1] 141.9687

Plot Predictions

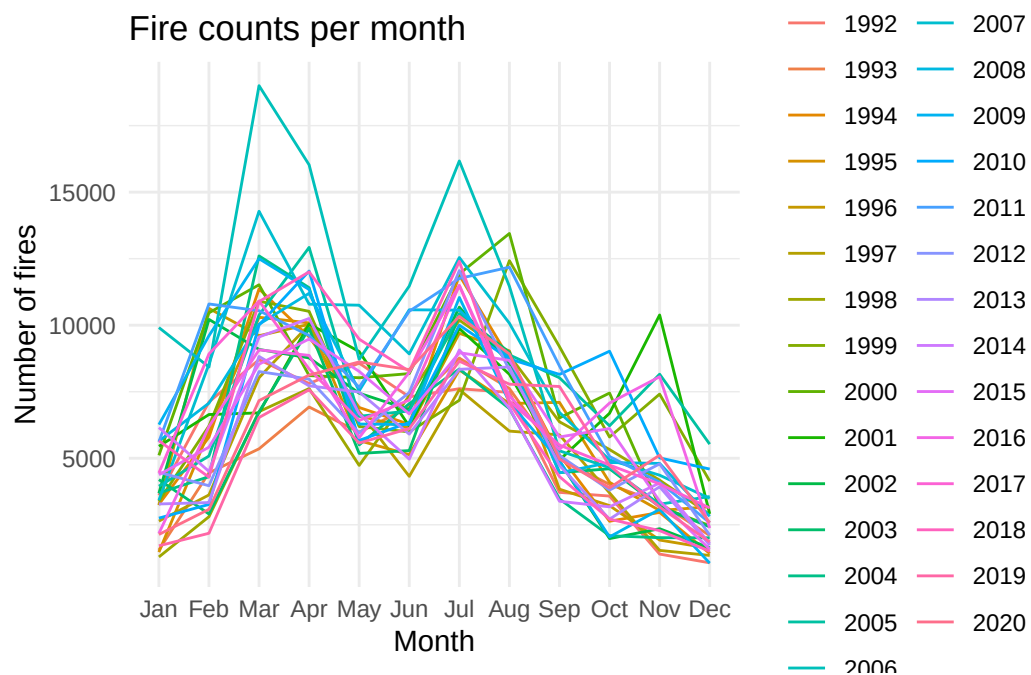


Conclusion

Appendix

Exploratory Analysis

```
ggplot(  
  fires_counts,  
  aes(x = month, y = n_fires, color = factor(year), group = year)  
) +  
  geom_line() +  
  labs(  
    title = "Fire counts per month",  
    x = "Month",  
    y = "Number of fires",  
    color = "Year"  
  ) +  
  theme_minimal()
```



```

ggplot(
  fires_month,
  aes(x = month, y = total_size, color = factor(year), group = year)
) +
  geom_line() +
  labs(
    title = "Total area burned per month",
    x = "Month",
    y = "Total FIRE_SIZE",
    color = "Year"
  ) +
  theme_minimal()

```

